Solvency Field Theory V9.0

Weak-Field Lag Action, Repaired Scalar Gravity, and the Quantitatively Constrained Goal 3 Bridge

Complete unified breakout draft up to the present closure point

Current logical status
Branch A falsified cleanly; Branch B survives as a partial kinematic/weak-field branch; Branch C* promoted and holding.
Goal 1 passed exactly: beta = 1.000 for repaired Branch C*.
Goal 2 passed exactly at first order: alpha_light = 4GM/(b c^2).
Goal 3 is not failed and not closed; it is narrowed to the local clock-field/source bridge.
Leading active bridge candidate: u^mu = grad^mu Phi / sqrt(-grad_nu Phi grad^nu Phi).
Latest quantitative stopping point: the phase bridge remains structurally viable, numerically healthy, and now bounded by recombination-scale mildness constraints on the composite bridge strength alpha*lambda.

Abstract. This document rewrites and unifies the earlier V8.7 weak-field lag-action closure note and the first V9.0 draft into one complete breakout paper. It records the full branch genealogy from Branch A through repaired Branch C*, the weak-field and post-Newtonian gates that were used to promote or reject each branch, and the present stopping point of the source-sector bridge program. The result is stronger than a metaphor and weaker than a finished nonlinear gravity theory. Branch A is falsified because it rescales the kinetic sector and produces a force that vanishes at rest. Branch B restores the Newtonian kinetic term, gravity at rest, and local special relativity in a uniform-n patch, but still fails the weak-field spatial metric. Branch C repairs that metric and closes the weak-field candidate sector. The repaired Branch C* line then survives the two classic scalar-gravity kill tests: the second-order PPN time-sector coefficient and the first-order Eddington light-deflection coefficient. Goal 3 remains open, but it is no longer vague. The remaining problem has been narrowed to a local clock-field/source bridge. The leading bridge candidate is a covariant phase-gradient congruence, which passes the FLRW limit, ledger-current divergence, and dust/radiation active-source checks. Subsequent bounded tests show that the bridge is not only structurally viable but numerically healthy under mild assumptions, and the latest recombination-scale bound tightens the composite bridge strength alpha*lambda without forcing absurd tuning. The paper therefore stops at the correct place: Goals 1 and 2 are closed, Goal 3 is quantitatively constrained rather than solved, and the next move is to derive the remaining bridge law from canonical SFT rather than continue unconstrained exploratory scanning.

1. Why this document exists

This paper exists because the weak-field lag-action thread outgrew the status of a side calculation. V8.7 preserved the first coherent weak-field lag-action closure through Branch C. The first V9 draft then added repaired Branch C*, Goal 1, Goal 2, and the initial Goal 3 bridge framing. The additional Goal 3A and Goal 3B calculations completed after that draft materially strengthened the document again. The clean solution is therefore not another patch note but a full unified rewrite.

The sequencing matters. The lag-action program did not begin as a demand for a brand-new theory of gravity. It began as an internal SFT question: if screening, compactness, maintenance cost, and observer-side filtering already behave like processor strain in the cosmological and strong-field sectors, can a local weak-field action be written that reproduces known matter motion and light propagation without breaking the rest of the framework?

The answer is now partly yes and partly not yet. A coherent weak-field / first-post-Newtonian candidate branch exists, and the two historical gates that typically destroy scalar candidates have been passed. But the source-sector bridge is still not derived from canonical V8 structure. This document records that exact state of play.

2. Canonical V8 background and why a local bridge is needed

The canonical V8 architecture is not itself a local covariant source theory. Its native count equation is an apparent-horizon balance over one Hubble time, and the constrained action presented there is explicitly minisuperspace. Within that architecture, the global ledger is real, the apparent-horizon settlement interval is real, and the matter contribution inside the native action is still the standard matter action for pressureless dust. This is why the present lag-action thread must be understood as a bridge program rather than as a trivial rewrite of V8.

The bridge problem is to connect the existing global ledger/count structure to a local weak-field action and source sector without smuggling in ad hoc machinery. The weak-field galactic program had already taught SFT that the relevant low-acceleration variable is not simply the strong-field compactness burden chi, but a weaker acceleration ratio tied to the derived scale a_SFT = cH0/(2pi). Meanwhile the strong-field sector had already locked compactness in as the correct variable for compact objects and relativistic regimes. The lag-action program arose from trying to understand whether a local 'processor strain' field could mediate between these sectors at the level of motion and lensing.

3. Logical program and gate structure

The thread was organized as a sequence of explicit gates rather than as a single global claim. Each branch was allowed to fail publicly, and any surviving branch had to earn promotion by passing the next gate rather than by accumulating metaphors.

Gate	Question	Status	Current answer
Branch history	Which local lag-action branch survives the elementary dyno tests?	Closed	A falsified; B partial; C* promoted and holding
Goal 1	Does the repaired branch pass the PPN beta gate?	Passed	beta = 1.000 exactly for C*
Goal 2	Does the repaired branch pass first-order light bending?	Passed	alpha = 4GM/(b c^2)
Goal 3	Can the source sector be closed so that radiation gravitates without a bad preferred frame?	Open	Narrowed to the local clock-field bridge and now quantitatively constrained

4. Branch history: what failed, what survived, and why

4.1 Branch A: the literal c/n substitution

Branch A implemented the cleanest possible reading of the original lag intuition. The free relativistic particle action was modified by inserting the local transmission speed $c_{local} = c / n$ directly into the spatial velocity term.

$$L_A = -m c^2 \sqrt{1 - n^2 v^2 / c^2}$$

This branch failed immediately. In a uniform background the low-velocity kinetic term becomes $(1/2) m n^2 v^2$ rather than $(1/2) m v^2$. When the lag field varies, the induced acceleration is proportional to $-(d \ln n / dx) v^2$, so a particle at rest does not fall. For a point-mass ansatz $n(r) = 1 + \alpha GM/(c^2 r)$, the resulting force is suppressed by v^2/c^2 and therefore vanishes in the Newtonian rest limit. Branch A is therefore falsified as a candidate local gravity action.

4.2 Branch B: promoting lag into the maintenance clock

Branch B repaired the core flaw in Branch A by allowing the lag field to slow the maintenance clock as well as the spatial transmission term.

$$L_B = -(m c^2 / n^2) \sqrt{1 - n^2 v^2 / c^2}$$

This is the first branch that became physically serious. In a uniform-n patch it can be rewritten as the standard free relativistic action in rescaled local coordinates, so local special relativity survives. In the weak-field limit the Newtonian kinetic term is restored exactly, and for $n(r) = 1 + \alpha GM/(c^2 r)$ the acceleration at rest is nonzero and inward. Choosing $\alpha = 1/2$ gives the correct Newtonian magnitude $-GM/r^2$.

Branch B nevertheless failed as a full relativistic weak-field theory because the associated spatial metric came out too soft. At first order it yielded $g_{tt} \sim= -(1 - 2U)$ while the spatial sector came out as $g_{rr} \sim= 1 - U$ rather than the standard weak-field form. Branch B therefore survives as a real partial kinematic and Newtonian-gravity sector, but it is not yet a satisfactory weak-field relativistic branch.

4.3 Branch C: metric repair

Branch C was introduced as the minimal metric repair of Branch B. Its job is not to preserve the prettiest metaphor; its job is to preserve the successful Newtonian and local-SR features of Branch B while hardening the spatial metric enough to reproduce the correct first-order relativistic geometry.

$$ds^2 = -(c^2 / n^4) dt^2 + n^4 d\ell^2$$

$$L_C = -(m c^2 / n^2) \sqrt{1 - n^8 v^2 / c^2}$$

In a uniform-n patch, Branch C again reduces to Minkowski spacetime after coordinate rescaling. In the weak field, taking $n(r) = 1 + \alpha GM/(c^2 r)$ yields $g_{tt} \sim= -(1 - 2U)$ and $g_{rr} \sim= 1 + 2U$ at first order when $\alpha = 1/2$, with $U = GM/(c^2 r)$. The Newtonian limit remains correct.

The price is conceptual rather than immediately physical. Branch C no longer preserves the literal $c_{local} = c/n$ interpretation. It should therefore be read as a metric-coupling repair branch: the lag field deforms time and space strongly enough to reproduce the correct first-order weak-field relativistic geometry.

5. Weak-field closure before the historical post-Newtonian gates

Before asking whether the repaired branch could survive the classic scalar-gravity graveyard, it was necessary to close its weak-field sector in the same disciplined way used elsewhere in SFT.

$$n(r) = 1 + GM / (2 c^2 r) = 1 + U/2$$

$$\nabla^2(n - 1) = - (2\pi G / c^2) \rho$$

For a point mass, the spherical vacuum solution outside the source is $n(r) = 1 + C/r$. Gauss-law matching fixes $C = GM/(2 c^2)$, reproducing the point-mass sourcing law exactly. At this stage Branch C had cleared the candidate-level weak-field tests: Newtonian mechanics in a uniform-n patch, Newtonian gravity at rest, local SR in a uniform-n inertial patch, the correct first-order weak-field metric, and the matching weak-field scalar sourcing law.

That was already substantial progress. It meant the lag-action thread had produced a coherent weak-field candidate sector rather than a loose interpretive sketch. But it still had to survive the classic post-Newtonian gates that historically destroyed earlier scalar gravity attempts.

6. Historical scalar-gravity context and why the next gates mattered

The next gates were framed sharply for historical reasons. Einstein's 1911 pre-GR light-bending calculation predicts $\alpha = 2GM/(b c^2)$, exactly half the observed value. Nordstrom's 1913 scalar theory predicts no light bending at all. Later scalar-tensor theories survive only by retaining a tensor sector and forcing the scalar contribution to hide in the solar system. The repaired Branch C* line therefore had to pass the same classic gates: the second-order PPN time-sector coefficient and the first-order Eddington light-deflection coefficient.

A further historical point matters here. The repaired Branch C* metric $g_{tt} = -\exp(-2U)$, $g_{rr} = \exp(2U)$ is structurally similar to the exponential-metric gravity associated with Yilmaz; the present thread does not arrive at that form by postulating an alternative metric a priori, but by forced coefficient matching inside the lag-action program after the linear Branch C closure failed the beta gate.

This historical framing is not ornament. It is the reason the lag-action thread matters. If the repaired branch had failed these gates, the honest conclusion would have been that SFT had reached a useful but nonviable scalar candidate. Passing them does not prove final success, but it changes the burden of proof: the active branch is no longer in the immediate graveyard of early scalar theories.

7. Goal 1: the PPN beta gate and the emergence of repaired Branch C*

The linear weak-field closure of Branch C does not survive the second-order time-sector test. With $n = 1 + U/2$ and $g_{tt} = -n^{-4}$, the expansion is:

$$g_{tt} = -(1 + U/2)^{-4} = -(1 - 2U + 5U^2/2 + \ldots)$$

$$g_{tt} = -(1 - 2U + 2\beta U^2 + \ldots)$$

Matching coefficients gives $\beta = 5/4$. That is not a near miss. It is a real failure of the linear weak-field closure at post-Newtonian order. But the failure is productive because it isolates the exact missing term.

$$n(U) = 1 + U/2 + a U^2 + \ldots$$

The beta coefficient then becomes $\beta = 5/4 - 2a$, so requiring $\beta = 1$ fixes $a = 1/8$ exactly. That coefficient is the second-order term of the exponential completion $n = \exp(U/2)$. This is the origin of repaired Branch C*.

$$\chi \equiv \ln n$$

$$\chi = U/2$$

$$n = e^{\chi} = \exp(U/2)$$

$$g_{tt} = -\exp(-2U), \quad g_{rr} = \exp(2U)$$

Now the second-order expansion of g_{tt} is

$$g_{tt} = -\exp(-2U) = -(1 - 2U + 2U^2 + \ldots)$$

and therefore $\beta = 1.000$ exactly. Repaired Branch C* clears Goal 1 symbolically, not numerically by tuning but by exact coefficient matching.

8. Goal 2: light deflection and the Eddington gate

Once Goal 1 was passed, the next gate was the first-order solar-system light-bending test. For the repaired metric in static isotropic form,

$$A(r) = \exp(-2U), \quad B(r) = \exp(2U)$$

$$n_{opt} = \sqrt{B/A} = \exp(2U) \approx 1 + 2U$$

Taking the straight-line first-order Born approximation for a null ray with impact parameter b,

$$\alpha = \int [- \partial(n_{opt} - 1) / \partial b] \, dz$$

$$n_{opt} - 1 \approx 2GM / (c^2 \sqrt{b^2 + z^2})$$

$$\alpha = 4GM / (b c^2)$$

So repaired Branch C* predicts the full Eddington/GR coefficient of 4 in units of GM/(b c^2). That is the second historical kill test cleared by the repaired branch. At first post-Newtonian order, the active branch is no longer behaving like the classic scalar theories that were historically discarded immediately.

9. Goal 3: source-sector closure and the remaining bridge problem

Goals 1 and 2 do not by themselves close the thread, because an action principle is not complete until the source sector is under control. The remaining problem is the electromagnetic/radiation source question: can the branch source gravity from radiation without collapsing either into trace-only Nordstrom behavior or into a bad preferred-frame energy coupling?

The strongest scalar target found in the search is the active-mass source

$$\sigma = (2/c^2) [(T_{\mu\nu} - 1/2 T g_{\mu\nu}) u^\mu u^\nu]$$

(equivalently, in c = 1 fluid units, $\sigma = \rho + 3p$)

This is strong because it gives the correct limits: dust sources as rho, while radiation with $p = \rho/3$ sources as 2rho rather than vanishing. But the source is only as good as the clock/congruence u^μ that enters it. The search therefore shifted from 'find a scalar source' to 'find a legitimate local clock field'.

Several shortcuts were rejected along the way. A raw current-normalization rule $u^\mu = J^\mu / |J|$ fails in null-radiation configurations. A naked preferred-frame T00 source is dangerous because it can violate the logic of relativity. A pure trace source dies immediately for radiation. The remaining live bridge is the phase-gradient candidate.

9.1 Phase-gradient clock field

$$u^\mu = \nabla^\mu \Phi / \sqrt{-\nabla_\nu \Phi \nabla^\nu \Phi}$$

Here Phi(x) is interpreted as an inherited macroscopic ledger phase, and the local clock/congruence is the normalized timelike phase gradient. This is the first Goal 3 bridge that is simultaneously covariant, mathematically nontrivial, and compatible with both matter and radiation sourcing.

9.2 Three bridge tests

1. Test 1: FLRW sanity check. In a homogeneous background with Phi = Phi(t), the phase gradient is purely timelike. As long as Phidot != 0, the normalized gradient reduces to the standard comoving cosmic flow $u^\mu = (1, 0, 0, 0)$ in comoving coordinates.
2. Test 2: ledger-current divergence. Define the ledger current relative to the phase clock as $J^\mu = -T^{\mu\nu} u_\nu / c^2$. Stress-energy conservation then gives $\nabla_\mu J^\mu = -(1/c^2) T^{\mu\nu} \nabla_\mu u_\nu$. For a perfect fluid in c = 1 units this specializes to $J^\mu = \rho u^\mu$ and $\nabla_\mu J^\mu = -p \theta$, with theta =

$\nabla_\mu u^\mu$. This is the cleanest precise form yet of the old 'ledger friction' intuition: dust gives zero divergence, while radiation gives a pressure/expansion work term against a relaxing phase background.

3. Test 3: radiation source law. In the local phase frame, take the perfect-fluid form $T_{\mu\nu} = (\rho + p) u_\mu u_\nu + p g_{\mu\nu}$, with trace $T = -\rho + 3p$. Then the active-mass scalar candidate is $\sigma = (2/c^2)(T_{\mu\nu} - 1/2\, T g_{\mu\nu}) u^\mu u^\nu$. In $c = 1$ fluid units, using $u^\mu u_\mu = -1$, one has $T_{\mu\nu} u^\mu u^\nu = \rho$ and $g_{\mu\nu} u^\mu u^\nu = -1$, so $\sigma = 2[\rho - 1/2\, T(-1)] = 2[\rho + 1/2(-\rho + 3p)] = \rho + 3p$. Therefore dust ($p = 0$) sources as $\sigma = \rho$, while radiation ($p = \rho/3$) sources as $\sigma = 2\rho$. The bridge therefore does not collapse back into the old trace-only scalar graveyard.

10. Goal 3B: backward triangulation and quantitative tightening

After the phase-gradient bridge cleared its structural checks, the next move was not to claim final closure but to ask whether the observed universe constrains the bridge tightly enough to work backwards. This turned Goal 3 into a three-constraint triangulation problem.

Constraint	What it fixes	Status
Goal 3A local clock structure	Requires the FLRW limit and the phase-gradient bridge identities to hold	Passed
CMB upper-bound logic	Constrains a transfer-weighted $k^2\, \delta\Phi_k / (a^2 \dot{\Phi})$ combination, not $\delta\Phi$ directly	Consistent but not unique
V8.6 observer-side closure	Constrains low-z path-averaged $U_{screen} = 1 - F$ along photon trajectories	Already closed at shell-model level

10.1 Minimal alpha-bridge

The safest first bridge family is not $F = \Phi$ identically, but a functional link between the cosmology-side screening variable $U_{screen} = 1 - F$ and the weak-field lag variable $\chi = \ln n$. In the ultra-weak regime the mildest choice is

$$U_{screen} \approx \alpha \chi$$

To connect the phase bridge to the weak-field variable, the mildest extra perturbative law is

$$\delta\chi = \lambda\, \delta\Phi$$

This does not close Goal 3 by itself. It only asks whether the weakest plausible bridge family is already numerically absurd. It is not.

10.2 CMB-side perturbative structure

With $\Phi(t,x) = \bar{\Phi}(t) + \delta\Phi(t,x)$, the phase-gradient clock in flat FLRW yields, to leading order,

$$u^0 \approx 1, \quad u^i \approx -(1 / (a^2 \dot{\Phi}_0))\, \partial^i \delta\Phi$$

$$\theta \approx 3H - (1 / (a^2 \dot{\Phi}_0))\, \nabla^2 \delta\Phi$$

$$\delta(\mathrm{div}\, J) \approx (p / (a^2 \dot{\Phi}_0))\, \nabla^2 \delta\Phi$$

In Fourier space, with $\nabla^2 \to -k^2$, this becomes

$$\delta(\mathrm{div}\, J)_k \approx \delta\Phi_k\, k^2\, p / (\dot{\Phi}_0\, a^2)$$

The structural consequence is important: the observed CMB does not constrain delta Phi directly. It constrains a transfer-weighted version of k^2 delta Phi_k / (a^2 Phi0dot). This is why Goal 3B becomes a constrained underdetermination problem rather than a direct inversion.

10.3 Minimal lambda-bridge scan

The first bounded scan eliminated the possibility that the bridge is only viable through absurd tuning. The relation

delta U_screen = (alpha * lambda * R) * (Delta T / T)

R = a^2 Phi0dot / (K_cmb * p * k^2)

With Delta T / T = 10^-5, target deltaU_screen = 10^-5, and mild exploratory ranges alpha in [0.1, 10], lambda in [0.1, 10], R in [0.1, 10], the scan found 16,211 matching grid points. One representative point was alpha = 0.1, lambda = 1, R = 10, giving the exact required ratio.

That result is important because it shows the bridge is not merely alive in principle; it is numerically healthy under order-of-magnitude assumptions.

10.4 FLRW clock law and thermal response estimate

The next bounded move was to attack the missing factor R from both sides. The mildest SFT-compatible background choice is to take the inherited phase as the accumulated e-fold count,

Phi_bar(t) = N(t) = ln a(t)

For radiation, rho_gamma is proportional to T^4, so a small temperature fluctuation satisfies Delta T / T = (1/4) delta rho_gamma / rho_gamma; this is the origin of the factor of 4. If ledger friction acts like a local energy injection or extraction Q over one Hubble time H^-1, then the mildest thermodynamic estimate is delta rho_gamma / rho_gamma ≈ Q / (H rho_gamma), so Delta T / T ≈ Q / (4 H rho_gamma). Identifying Q = delta(div J)_k and comparing with Delta T / T = K_cmb * delta(div J)_k gives

K_cmb ≈ 1 / (4 H rho_gamma)

K_cmb * p ≈ 1 / (12 H)

Delta T / T ≈ k^2 delta Phi_k / (12 a^2 H Phi0dot)

Delta T / T ≈ k^2 delta Phi_k / (12 a^2 H^2) if Phi0dot = H

delta Phi_k ≈ 12 (aH/k)^2 (Delta T / T)

R ≈ 12 (aH/k)^2

This is the first clean non-arbitrary estimate of R. It says the missing factor is not a free mystery constant but roughly a horizon-scale ratio. Near the horizon scale at recombination, k ~ aH and therefore R ~ 12, which is comfortably mild in the sense of the previous scan. The team favorite option Phidot ~= H and the mild thermal-response estimate are therefore mutually compatible.

10.5 Recombination-scale mode sanity check

Using Delta T / T = 10^-5 and the relation delta Phi_k ≈ 12 (aH/k)^2 (Delta T / T), the recombination-scale mode scan gives the following nominal perturbation amplitudes for x = k/(aH).

x = k/(aH)	deltaPhi_k	deltaU_nominal (alpha=lambda=1)	deltaU_range (alpha,lambda in [0.1,10])
0.5	4.8e-4	4.8e-4	[4.8e-6, 4.8e-2]

1	1.2e-4	1.2e-4	[1.2e-6, 1.2e-2]
2	3.0e-5	3.0e-5	[3.0e-7, 3.0e-3]
3	1.333e-5	1.333e-5	[1.333e-7, 1.333e-3]
5	4.8e-6	4.8e-6	[4.8e-8, 4.8e-4]
10	1.2e-6	1.2e-6	[1.2e-8, 1.2e-4]

This is good news for the bridge. Horizon-scale modes (x ~ 1) remain at the 10^-4 level, and sub-horizon modes become smaller quickly. The bridge does not require wild relic-phase structure merely to reproduce the observed 10^-5-level CMB anisotropy.

10.6 alpha*lambda mildness bounds: the present quantitative stopping point

The final narrowing pass converted the recombination-mode scan directly into bounds on the composite bridge strength alpha*lambda by imposing simple mildness thresholds on deltaU_screen.

Mode cut	Mildness threshold	Controlling mode	Max(alpha*lambda)
x >= 1	deltaU <= 1e-4	x = 1	0.833
x >= 1	deltaU <= 1e-3	x = 1	8.333
x >= 0.5	deltaU <= 1e-4	x = 0.5	0.208
x >= 0.5	deltaU <= 1e-3	x = 0.5	2.083

This is the first true quantitative tightening of Goal 3. For the physically important horizon-scale and sub-horizon recombination modes, keeping the bridge very mild at deltaU <= 10^-4 requires alpha*lambda $\lesssim$ 0.833. The looser deltaU <= 10^-3 threshold allows alpha*lambda $\lesssim$ 8.33. Including the x = 0.5 super-horizon-ish test point tightens the stricter bound to alpha*lambda $\lesssim$ 0.208.

That is precisely where the program should stop for now. The bridge is still alive and still numerically plausible, but it is no longer unconstrained. Further progress now requires a real derivation of the missing law from canonical SFT rather than more exploratory scans.

11. Present status and exact stopping point

The lag-action thread has now reached a clear and defensible stopping point. The weak-field branch is not just alive; it is the first scalar candidate in the program to survive the classic historical gates that usually kill such theories immediately. The source-sector bridge is not closed, but it is also not vague. It is structurally viable, numerically healthy, and now quantitatively bounded.

The exact status statement of V9 should therefore be: Goals 1 and 2 are closed. Goal 3 is not failed and not solved. It has been reduced to a sharply constrained local clock-field bridge whose leading active candidate is the phase-gradient congruence u^mu proportional to grad^mu Phi. The next move is to derive the remaining bridge law from canonical SFT rather than to continue open-ended exploratory fitting.

12. Conclusion

The weak-field lag-action program began as a tempting intuition and could easily have stayed there. What changed is that the wrong versions were forced to fail in public. Branch A died because it mistook a kinematic slogan for a viable action. Branch B kept the Newtonian and local-SR pieces alive but still bent the spatial geometry the wrong way. Branch C repaired the weak-field metric. Branch C* then survived the two classic post-Newtonian gates that historically destroy scalar candidates.

Goal 3 remains the unfinished bridge, but it is now unfinished in the right way. The phase-gradient clock field gave the first mathematically legitimate origin for the local congruence. The backward-triangulation program then showed that the bridge is consistent with both the observer-side screening closure and the CMB-scale perturbation architecture. The latest mildness bound finally tightened the composite bridge strength alpha*lambda enough that the theory now has an exact quantitative stopping point.

That is a real result. The program is no longer asking whether a weak-field lag-action sector might exist. It now has one. The next job is not to guess harder; it is to derive the remaining law that would make the bridge canonical.

Appendix A. Branch genealogy and dyno outcomes

Branch	Action / metric	What it fixed	What it failed	Status
A	L_A = -m c^2 sqrt(1 - n^2 v^2 / c^2)	Clean literal c/n picture	Kinetic term rescales by n^2; force vanishes at rest; Newtonian gravity fails	Rejected
B	L_B = -(m c^2 / n^2) sqrt(1 - n^2 v^2 / c^2)	Exact Newtonian kinetic term; gravity at rest; local SR in uniform n	Weak-field spatial metric too soft (g_rr ~= 1 - U)	Partial sector
C	ds^2 = -(c^2 / n^4) dt^2 + n^4 dℓ^2	Keeps Branch B gains and repairs first-order weak-field metric	Linear closure fails Goal 1 at beta = 5/4	Promoted then repaired
C*	g_tt = -exp(-2U), g_rr = exp(2U)	Passes Goal 1 and Goal 2 exactly; supports Goal 3 bridge program	Remaining source-sector bridge still open	Promoted and holding

Appendix B. Minimal equation set at the present stopping point

chi = ln n

n = exp(U/2)

g_tt = -exp(-2U), g_rr = exp(2U)

alpha_light = 4GM / (b c^2)

Phi_bar(t) = ln a(t)

Phi0dot = H

sigma = (2/c^2)[(T_mu_nu - 1/2 T g_mu_nu) u^mu u^nu]

u^mu = ∇^mu Phi / sqrt(-∇_nu Phi ∇^nu Phi)

$$\delta(\mathrm{div}\, J)_k \approx \delta\Phi_k \cdot k^2 \cdot p / (\Phi_0\dot{} \cdot a^2)$$

$$\delta\Phi_k \approx 12\, (aH/k)^2\, (\Delta T / T)$$

$$\delta U_{screen} \approx (\alpha \cdot \lambda)\, \delta\Phi_k$$

Appendix C. Quantitative bridge checkpoints

The bounded quantitative checkpoints reached in Goal 3B can be summarized compactly as follows.

- Minimal lambda-bridge health scan: with $\Delta T / T = 10^{-5}$, target $\delta U_{screen} = 10^{-5}$, and alpha, lambda, R each allowed to vary over [0.1, 10], the scan found 16,211 matching grid points. The bridge is therefore not numerically forced.
- Mild FLRW clock law: the choice $\Phi_{bar} = \ln a$ implies $\Phi_0\dot{} = H$ and is the least invasive background clock identification consistent with the FLRW sector.
- Crude thermodynamic response estimate: $K_{cmb} \sim= 1 / (4 H \rho_\gamma)$, which yields $R \sim= 12(aH/k)^2$ and therefore keeps the missing factor in a horizon-scale regime rather than an arbitrary constant regime.
- Recombination mode sanity check: for $x = k/(aH) = 1$, the nominal inferred phase perturbation is $\delta\Phi_k \sim= 1.2e-4$; for x = 2, 3, 5, and 10 the inferred perturbation drops to 3.0e-5, 1.33e-5, 4.8e-6, and 1.2e-6 respectively.
- Mildness bound on the composite bridge strength: if one demands $\delta U_{screen} <= 1e-4$ on horizon-scale and sub-horizon recombination modes, then $\alpha\lambda <= 0.833$. Including the looser x = 0.5 test point tightens that to $\alpha\lambda <= 0.208$.

These checkpoints do not close Goal 3, but they do define the exact quantitative edge of the present bridge program.